

A CLOSER LOOK

When sperm meets egg

This is the moment of the miracle: when a single sperm from the 200–500 million contenders in a single ejaculate penetrates a mature egg that only survives for 24 hours. The journey has taken the tiny sperm around 5–20 minutes, and most of its brothers have fallen by the wayside in the vital race.

1 This sperm has beaten millions of others to reach the egg in the fallopian tube. Only about 200–300 sperm made it this far. The others bind to the surface of the egg, but this winning sperm pushes through the egg's layers to penetrate it.

A long, whip-like tail propels the sperm on its swim toward the egg.

The midsection houses a spiral mitochondrion—it's the powerhouse that provides the sperm with energy.

The sperm head contains the male DNA—including the X or Y chromosome that will determine whether your baby is a girl or boy.

2 The sperm is able to enter the egg's corona radiata and zona pellucida layers with the help of enzymes in its acrosome—a caplike structure on its head. The acrosome will shed when the job is done, so the sperm can fuse with and fertilize the egg.

The egg is encased by a thick transparent double membrane called a zona pellucida.

